

Low-Rank Structure Is Geometry

Se Eun Choi

Beginner worksheet · 30–40 minutes · no proofs or code from scratch

Open the named Pluto Lab, use only the relevant experiment control, inspect the visual result, and answer briefly. The interactive version is `notebooks/05_ExerciseSheet.jl`. The themes are adapted at an introductory level from Paul Breiding's Tensorlab exercises; the questions are rederived for TensorKitchen and are not verbatim copies.

The goal is to distinguish the represented object, its coordinates, the model assumption, and the evidence needed before interpreting components.

Exercise 1 — Predict a multilinear map

`notebooks/00_Primer.jl` · Section 1 · about 6 minutes

Let T have size $(4, 3, 3)$, with $M1$ of size $(2, 4)$ and $M2, M3$ of size $(2, 3)$. Predict before running the multilinear mode-product experiment.

1. Predict the size of $Y = T \times_1 M1 \times_2 M2 \times_3 M3$.

☐ $(2, 2, 2)$ ☐ $(4, 2, 2)$ ☐ $(2, 3, 3)$ ☐ $(4, 3, 3)$

2. Predict the intermediate sizes after the mode-1 map and then the mode-2 map.

3. Run the experiment. What final size does the notebook report?

4. Which original dimension is transformed independently by each of $M1$, $M2$, and $M3$?

5. How many entries does Y contain?

☐ 6 ☐ 8 ☐ 12 ☐ 18

6. What does this experiment show about a tensor as a multilinear object?

Exercise 2 — Build, decompose, reconstruct

notebooks/00_Primer.jl · Section 3 · about 7 minutes

Generate a rank-3 tensor from A of size (3, 3), B of size (3, 3), and C of size (2, 3). Fit a rank-3 CPD and inspect the returned object.

1. What is the size of the generated tensor, and how many rank-one terms were used?

2. What is the format of the returned CP decomposition?

3. Reconstruct $\hat{T}$. What size does it have?

4. Record $\|T - \hat{T}\|_F / \|T\|_F$ from the notebook.

5. If the relative error is nearly zero, what has been verified?

6. What has not been verified by a near-zero error?

7. Must the fitted factor columns equal the original columns entry by entry?

☐ Yes ☐ No

Exercise 3 — Same object, different coordinates

notebooks/01_OneObjectManyCoordinates.jl · Sections 1A–1B · about 8 minutes

Open the Matrix gauge experiment. Make a prediction, reveal the experiment, and then move the gauge slider.

Measure

1. Set $\log_{10}(s) = 0$. Record s and $\kappa(Q)$.

2. Set $\log_{10}(s) = 3$. Record s and $\kappa(Q)$.

3. Tick every quantity that changes strongly.

☐ factor coordinates ☐ factor conditioning ☐ represented matrix

4. In the CP rescaling and permutation experiment, does the represented tensor change?

☐ Yes ☐ No

Predict and explain

5. For $Q(s) = \text{diag}(s, s^{-1})$, what do you expect as s becomes very large?

☐ X changes dramatically ☐ the factors may become badly scaled
☐ the reconstruction error stays near zero ☐ $\kappa(Q)$ increases

6. If $A' = AQ$ and $B' = BQ^T$, complete $A'B'^T = \underline{\hspace{2cm}}$.

7. Two different factor pairs produce the same matrix. Which object should we regard as the underlying object?

☐ A ☐ B ☐ the pair (A, B) ☐ the represented matrix X

8. If $X = A_1 B_1^T = A_2 B_2^T$, must $A_1 = A_2$ and $B_1 = B_2$?

☐ Yes ☐ No

9. A relative reconstruction error is 10^{-14} . Which conclusion is justified?

☐ the factors are unique ☐ the recovered matrix is extremely close
☐ the problem is well-conditioned ☐ every factor was recovered correctly

10. Finish the sentence: A low-rank factorization is a of the underlying matrix. The coordinates can change while the represented stays the same.

Exercise 4 — Multilinear rank and model choice

notebooks/02_GeometryAtlas.jl · HOSVD check and model cards · about 8 minutes

Compute the HOSVD summary of the supplied random tensor of size $(3, 3, 2)$, then compare CP, Tucker, and BTD on the two-block target.

Multilinear rank

1. Record the sizes of the three unfoldings.

2. Record their matrix ranks. What is the tensor's multilinear rank?

3. Why can the third multilinear rank never exceed 2 for a tensor of size $(3, 3, 2)$?

4. Can a tensor have CP rank 3 while its third multilinear rank is only 2?

☐ Yes ☐ No

Choose the structural model

5. Which model uses one shared list of rank-one components across all modes?

☐ CP ☐ Tucker ☐ BTD

6. Which model uses one core with a possibly different compression rank for each mode?

☐ CP ☐ Tucker ☐ BTD

7. The Lab 2 target is a sum of two small Tucker blocks. Which model most directly mirrors that construction?

☐ CP ☐ Tucker ☐ BTD

8. If one fitted model has the smallest error in this finite run, does that prove it is always the best model?

☐ Yes ☐ No

Exercise 5 — Observe stagnation, diagnose its geometry

notebooks/03_OptimizationFailureMuseum.jl · about 10 minutes

Use the failure map, collision-distance dial, three-case comparison, solver race, and plateau microscope to separate an observation from its possible cause.

Collision and the ALS Gram matrix

1. Set $\rho = 0.90$. What is the sign-invariant rank-one collision distance?

2. At $\rho = 0.90$, what are the off-diagonal entries of the two factor Gram matrices and of the ALS Gram matrix?

3. As ρ approaches 1, which eigenvalue becomes small: $1 + \rho^2$ or $1 - \rho^2$?

☐ $1 + \rho^2$ ☐ $1 - \rho^2$

4. In plain language, what becomes hard for ALS when the $[1, -1]$ direction nearly disappears?

5. Does this collision example require large opposite weights?

☐ Yes ☐ No

Trajectory and plateau diagnosis

6. Why must every solver in the race start from the same CPDPoint?

7. What observation triggers the plateau flag, and which diagnostics are used afterward to test a collision explanation?

8. The poor-start case is slow even though its component distance is large and its ALS condition number is moderate. Does slow optimization by itself prove collision?

☐ Yes ☐ No

9. Can a reconstruction error be small while individual components remain difficult to interpret reliably?

☐ Yes ☐ No

Exercise 6 — From activations to a concept claim

notebooks/04_NeuralRepresentations.jl · Concept explorer and evidence ladder · about 14 minutes

Audit one candidate concept in the synthetic CRAFT-style pipeline. Separate what the factorization shows from the semantic and behavioral claims that still require evidence.

Read the factorization

1. What does one row of the activation matrix A represent?

2. In $A \approx UW^T$, what do $U[i,j]$ and $W[:,j]$ represent?

3. Move the rank slider from 1 to 5. At which rank are the three planted ingredients easiest to name, and what happens on either side?

4. Choose one concept. Which evidence helps you propose a human label: the CAV alone or the high-usage crops?

☐ CAV alone ☐ High-usage crops

Audit the claim

5. In the perturbation experiment, is the most present concept necessarily the most important to the class score?

☐ Yes ☐ No

6. Why is nonnegativity not enough for a concept claim? Give one stability check.

7. Why is the sample $\times$ location $\times$ feature factorization in 4H not CRAFT?
